



Analog IC Design

CMOS Technology

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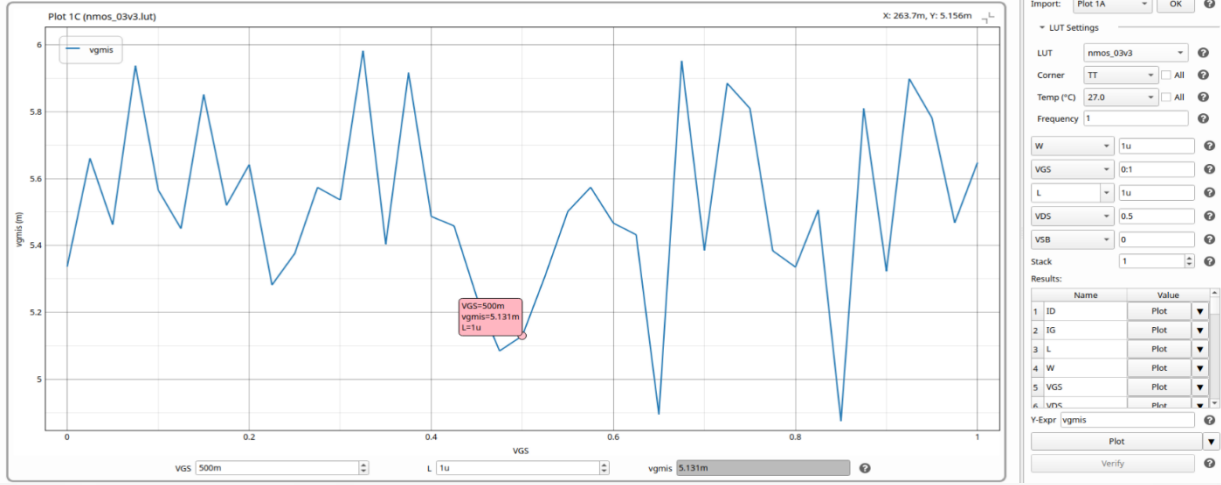
Section 3 - Lab 05

Current mirror (CM)

Sizing chart

We want sinking current source which is **NMOS** device.

The required $\lambda < 0.1$ per unit volt.



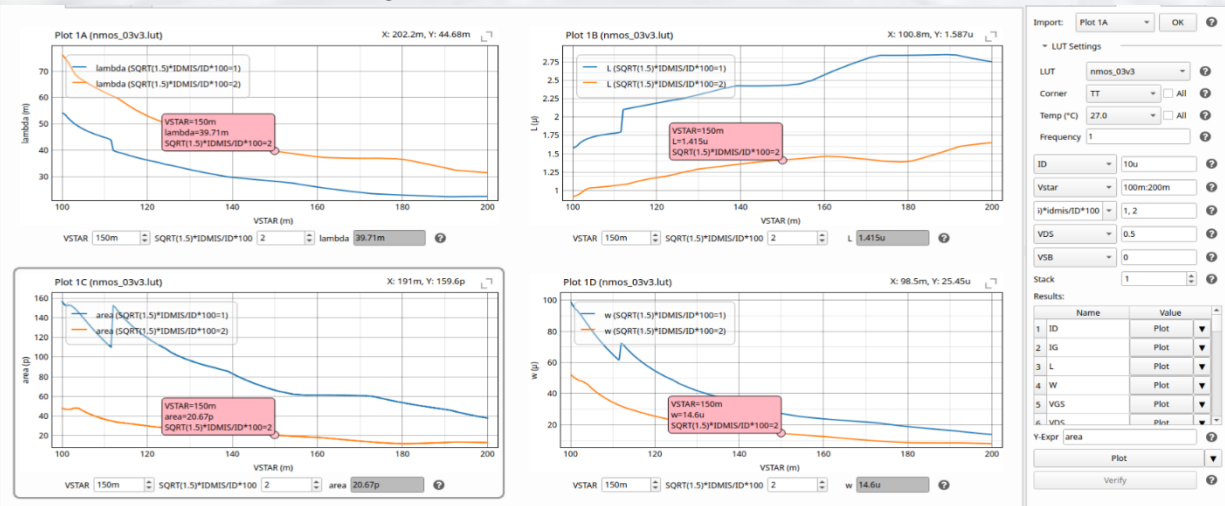
ADT mismatch (around 5.15mv)

```

46930 .lib fets_mm
46931 .subckt nfet_03v3 d g s b w=l-e-5 l=2.8e-7 nf=1 m=1 as=0 ad=0 ps=0 pd=0 nrd=0 nrs=0 par=1 dtemp=0 sa=0 sb=0 sd=0
46932 .param par_vth=0.007148
46933 .param par_k=0.007008
46934 .param par_l=1.5e-7
46935 .param par_w=1e-7
46936 .param par_leff='l-par_l'
46937 .param par_weff='par*(w-par_w)'
46938 .param p_sqrtarea='sqrt((par_leff)*(par_weff))'
46939 .param var_k='0.7071*par_k*1e-06/p_sqrtarea'
46940 .param mis_k='agauss(0,var_k,1)'
46941 .param var_vth='0.7071*par_vth*1e-06/p_sqrtarea'
46942 .param mis_vth='agauss(0,var_vth,1)'
46943 .m d g s b nfet_03v3 ad=ad as=as delvto='mis_vth*sw_stat_mismatch' l=l m=m mulu0='1-mis_k*sw_stat_mismatch' nf=nf nrd=nrd nrs=nrs pd=pd ps=ps sa=sa
46944 .ends nfet_03v3
    
```

From the model file, $\sigma_{VT} = \frac{A_{VT}}{\sqrt{WL}} = Var_V_{th} = 0.7071 \times 0.007148 \approx 5.05mv$ for $1\mu m^2$ area.

The value of A_{VT} getting from ADT > model file, since ADT takes all mismatches in calculation, but they approximately are equal because V_{TH} mismatch is the largest mismatch.



Sizing

As shown in the graphs above, as V^* increases the area needed decrease ($w \downarrow, L \uparrow$) that reduces the short channel effects and $\frac{g_m}{I_D}$, consequently reducing the drain current error.

So, the results show that the higher V^* is desirable as it provides small λ , $area$, and met the mismatch specs.

At $V^* = 150mv$ the design point satisfies the mismatch and λ constraints.

our sizing chart for ($I \rightarrow V$) transistors:

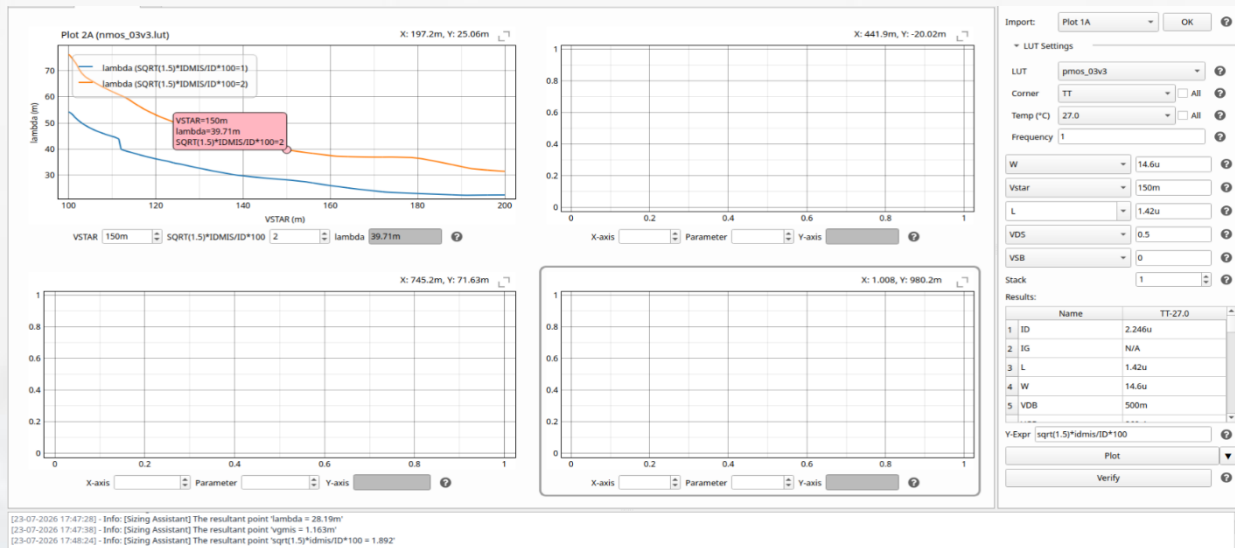
$$L = 1.42\mu m$$

$$W = 14.6\mu m$$

And to satisfy the mirroring ratio $m = 2$, the sizing of ($V \rightarrow I$) transistors will be as above with multiplier constant $m = 2$.

ID	10u	?
Vstar	150m	?
sqrt(1.5)*idmis/ID*100	2	?
VDS	0.5	?
VSB	0	?
Stack	1	?

Results:		TT-27.0
3	L	1.42u
4	W	14.6u

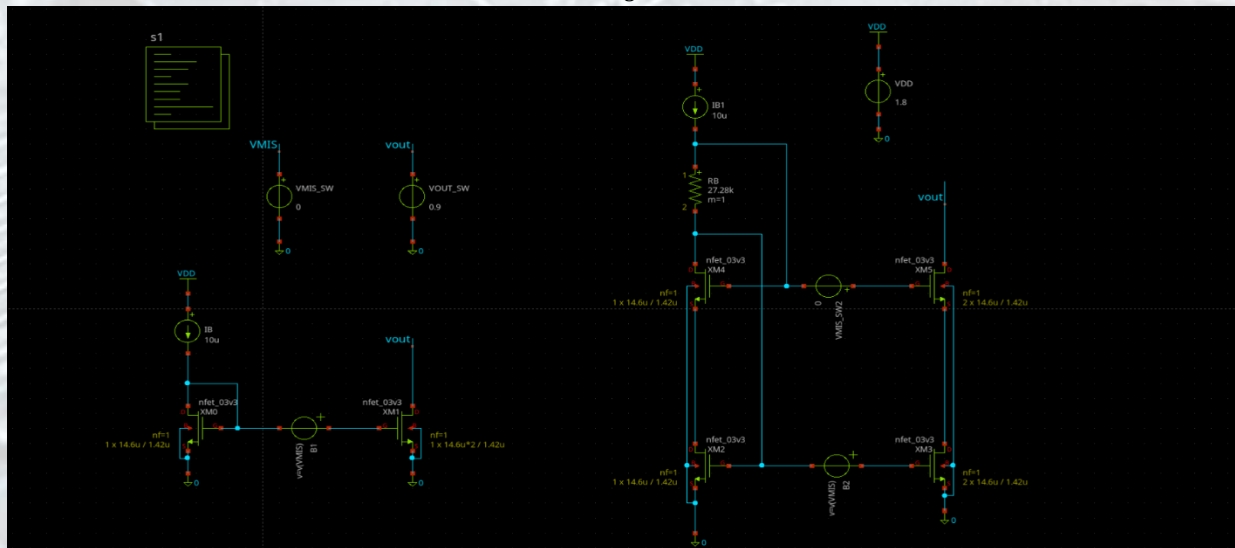


Double check on the specs at the selected sizing for $V^* = \text{compliance voltage} = 150mv$:

$$\lambda = 28.19mv < 0.1v, \frac{\sigma(I_{out})}{I_{out}} = 1.892\% < 2\%$$

$$\text{for the area} = 14.6\mu m \times 1.42\mu m \approx 20.7\mu m^2 V_{gmis} = 1.163mv.$$

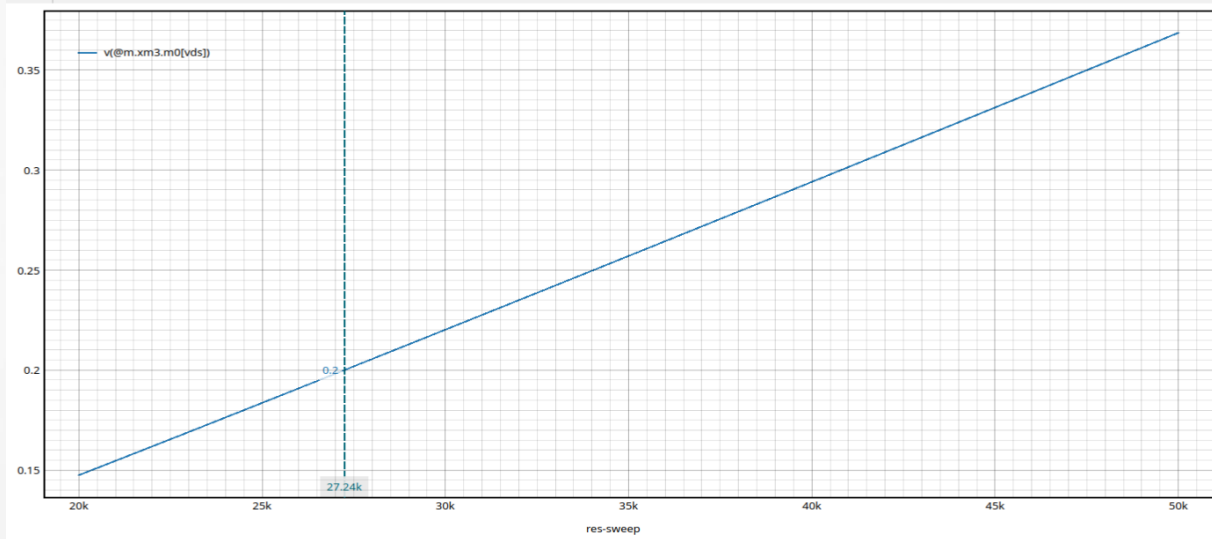
$$\text{for the area} = 2 \times 14.6\mu m \times 1.42\mu m \approx 41.4\mu m^2 V_{gmis} = 1.163mv \times \sqrt{2} = 1.645mv.$$



OP analysis

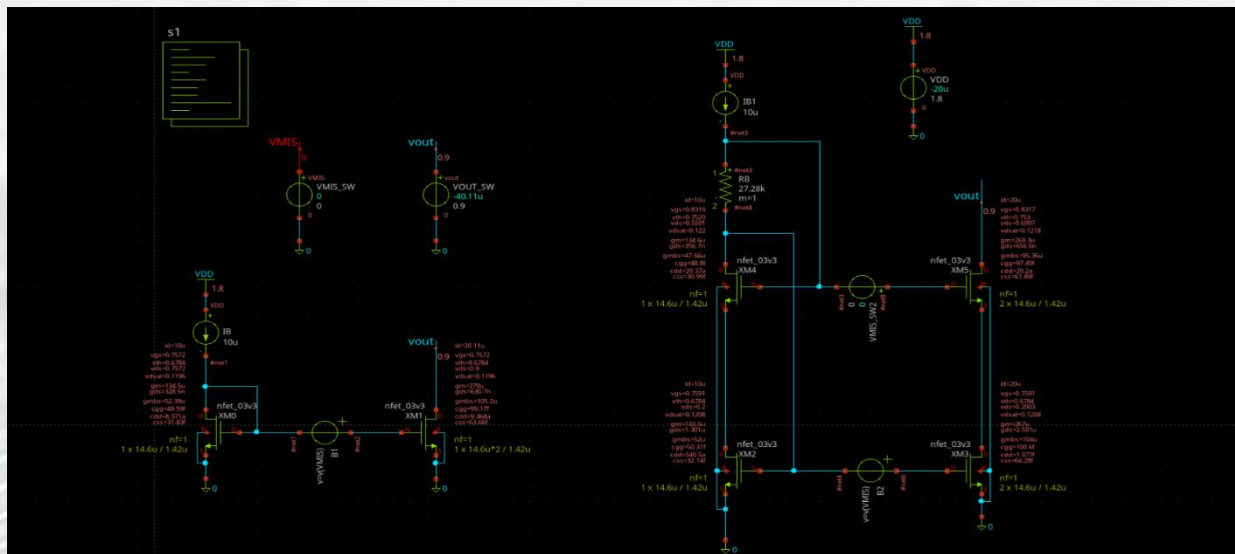
calculating a rough value for R_B

$$R_B = \frac{V_{GS4} + V_{DS2} - V_{GS2}}{I_B} \approx \frac{V_{DS2}}{I_B} = \frac{150\text{mV} + 50\text{mV}}{10\mu\text{A}} = 20\text{K}\Omega$$



V_{DS3} vs R_B

R_B that satisfy the 50mV saturation margin requirement is $\approx 27.5\text{K}\Omega$

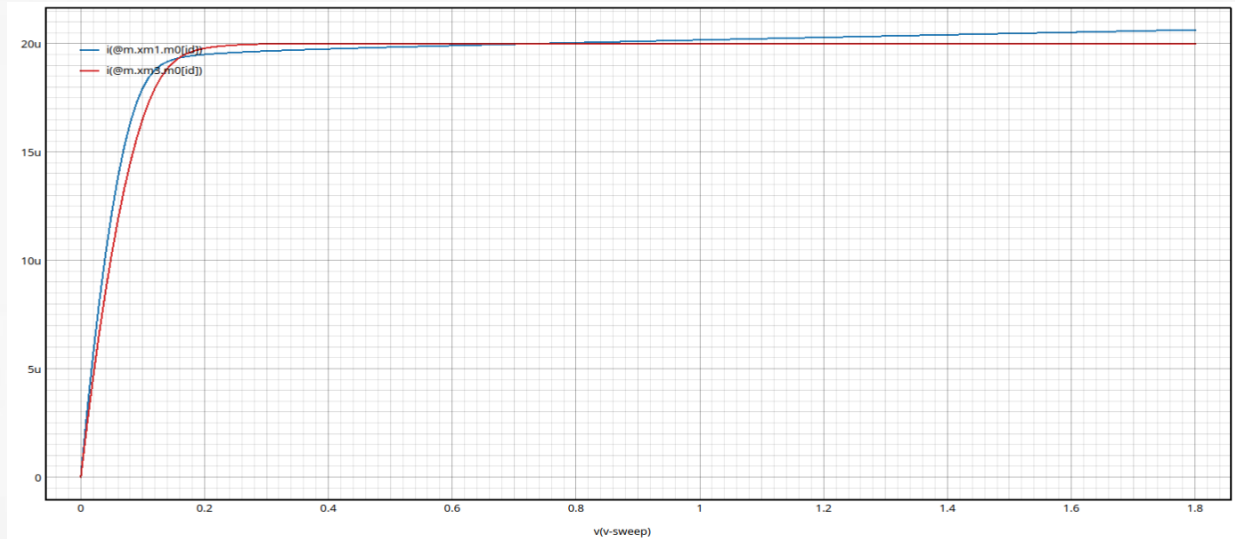


OP analysis point

Do all transistors operate in saturation?

Yes, all of them in saturation, since all have $(V_{GD} = V_{GS} - V_{DS}) < V_{TH}$

DC analysis



I_{out} : ■ wide swing CM, ■ simple CM

Comment:

For **wide swing CM**, the curve is **more ideal** and has very **low dependance on V_{DS}** but its saturation region starts at voltage larger than the simple CM. The estimate value of its **compliance voltage** is $\approx 0.25v$.

For **simple CM**, the curve is **less ideal** and has **some dependance on V_{DS}** but its saturation region starts at voltage lower than the wide swing CM. The estimate value of its **compliance voltage** is $\approx 0.15v$.

I_{out} of the simple CM is **exactly** equal to $I_B \times 2$ at only a **specific** value of V_{out} since the simple CM have some dependance on V_{DS} (I_{out} increase with as $V_{DS} \uparrow$), so I_{out} can't remain constant as $V_{DS} \uparrow$.

Calculate the percent change in I_{out} when V_{OUT} changes from 0.5V to 1.5V

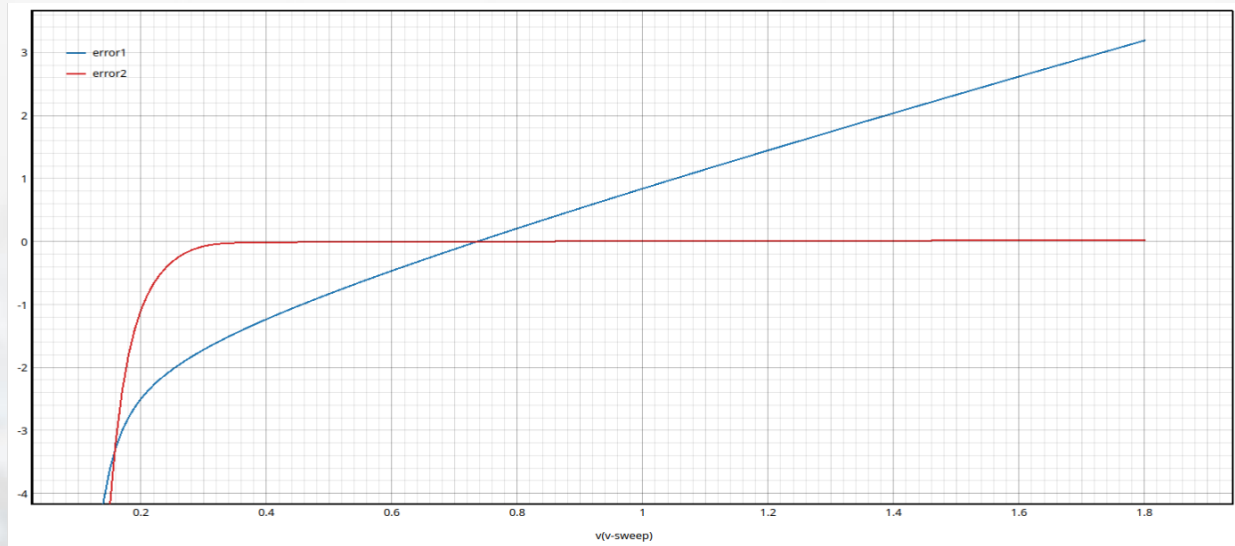
```
No. of Data Rows : 181
Simple Current Mirror Simulation Results
i1                = 1.983370e-05
i2                = 2.046595e-05
perc_change       = 3.137744e+00
Wide Swing Current Mirror Simulation Results
i1                = 1.999844e-05
i2                = 2.000320e-05
perc_change       = 2.379902e-02
```

From ADT: percent change $\approx \lambda = 33\text{mv} * 100\% \approx 3.3\%$ per one volt

Comment:

For the simple CM, the value is approximately matched the simulation result.

But for the wide swing CM the error is significantly reduced since the transistor placed in cascode circuit that has boosted R_{out} , $\Delta I_{out} = \frac{\Delta V_{out}}{R_{out}}$

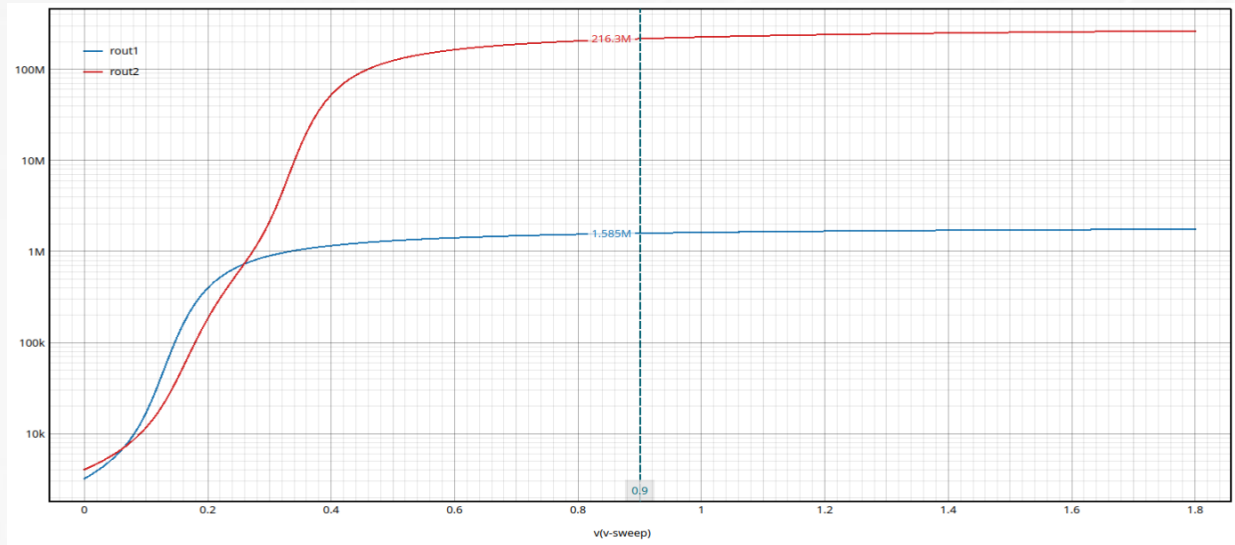


Percentage error: ■ wide swing CM, ■ simple CM

Comment:

In saturation region, the simple CM has variable error that depends on V_{DS} (the error increase as V_{DS} ↑), the wide swing CM has very small error that is almost constant (very low dependence on V_{DS}).

Out of saturation, both have very large error.



R_{out} : ■ wide swing CM, ■ simple CM

Comment:

The wide swing CM has very large output resistance $\approx 216.3M\Omega$ that reduces the change in the output current

$$\Delta I_{out} = \frac{\Delta V_{out}}{R_{out}}$$

The simple CM has very small output resistance relative to the wide swing CM $\approx 1.58M\Omega$ that cause large change in the output current relative to the wide swing CM.

Note: R_{out} change with V_{out} as r_o change, since R_{out} equals a function of r_o .

Analytically calculate Rout of both circuits at $V_{OUT} = \frac{V_{DD}}{2} = 0.9v$.

$$\text{For wide swing CM: } R_{out} \approx r_{o3}(g'_{m5}r_{o5}) = \frac{g_{m5} + g_{mb5}}{g_{ds5}g_{ds3}} = \frac{269.3\mu S + 95.36\mu S}{656.6nS \cdot 2.591\mu S} \approx 214.35M\Omega$$

$$\text{For simple CM: } R_{out} \approx r_{o1} = \frac{1}{g_{ds1}} = \frac{1}{630.7nS} \approx 1.586M\Omega$$

parameter	Simulation	Analytical	Error
Wide swing CM R_{out}	216.3M Ω	214.35M Ω	1.95M $\Omega \equiv 0.9\%$
Simple CM R_{out}	1.585M Ω	1.586M Ω	1K $\Omega \equiv 0.063\%$

Mismatch

For simple CM:

The % change in $I_{out} \approx 2.27\%$

For wide swing CM:

The % change in $I_{out} \approx 2.24\%$

```
No. of Data Rows : 18
Simple Current Mirror Simulation Results
i1      = 2.010552e-05 at= 0.000000e+00
i2      = 2.056807e-05 at= 1.700000e-03
perc_change = 2.274449e+00
Wide Swing Current Mirror Simulation Results
i1      = 2.000068e-05 at= 0.000000e+00
i2      = 2.045463e-05 at= 1.700000e-03
perc_change = 2.244205e+00
```

Analytically calculate the percent change in I_{out}

For simple CM:

$$= \frac{g_{m1}V_{gms}}{I_{out}} \times 100\% = \frac{270\mu S \cdot 1.645mv}{20\mu A} \times 100\% \approx 2.22\%$$

For wide swing CM:

$$\frac{g_{m3}V_{gms}}{I_{out}} \times 100\% = \frac{267\mu S \cdot 1.645mv}{20\mu A} \times 100\% \approx 2.2\%$$

parameter	Simulation	Analytical	Error
Wide swing CM $\Delta I_{out}\%$	2.24%	2.2%	0.04
Simple CM $\Delta I_{out}\%$	2.27%	2.22%	0.05

For cascode wide swing CM:

The % change in $I_{out} \approx 0.016\%$

```
No. of Data Rows : 18
Wide Swing Current Mirror Simulation Results (Cascode)
i1      = 2.000068e-05 at= 0.000000e+00
i2      = 2.000387e-05 at= 1.700000e-03
perc_change = 1.594819e-02
```

Analytically calculate the percent change in I_{out}

$$= \frac{G_m V_{gms}}{I_{out}} \times 100\%, G_m = \frac{g_{m5} + g_{mb5}}{1 + (g_{m5} + g_{mb5})R_s} \approx \frac{1}{r_{o3}} = g_{ds3} = 2.6\mu S \rightarrow = \frac{2.6\mu S \cdot 1.645mv}{20\mu A} \times 100\% \approx 0.02\%$$

parameter	Simulation	Analytical	Error
Cascode CM $\Delta I_{out}\%$	0.016%	0.02%	0.004

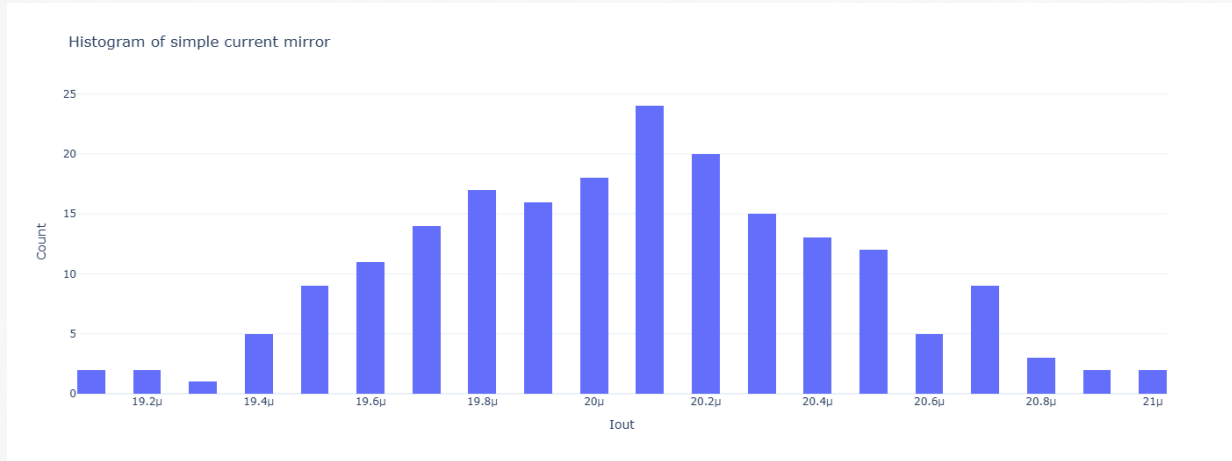
Which mismatch contribution is more pronounced? Why?

V_{TH} is more pronounced as V_{ov} is low voltage since, $\frac{\Delta I}{I} \approx \frac{\Delta(W/L)}{W/L} + \frac{2\Delta\beta}{\beta} - \frac{2\Delta V_{TH}}{V_{ov}}$

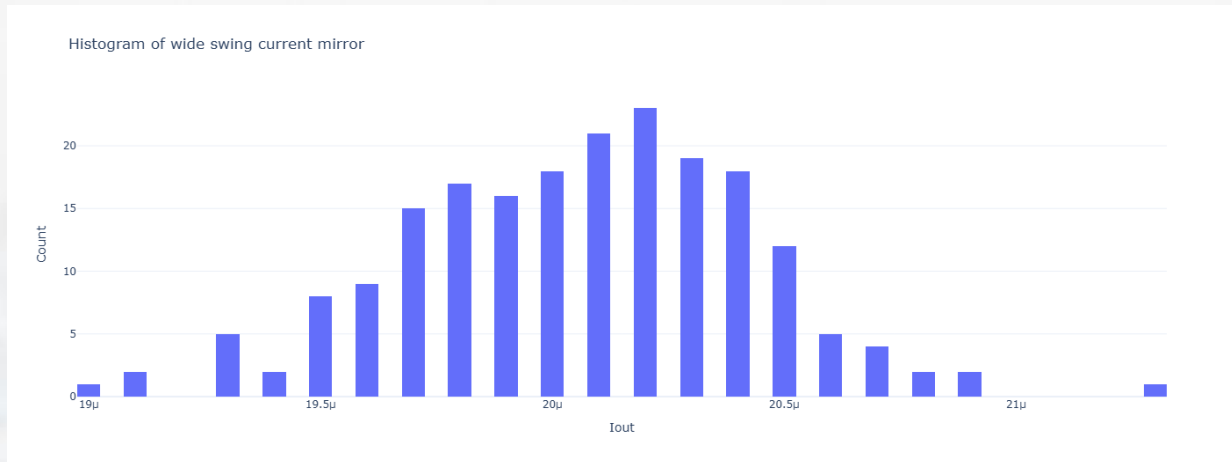
For a given total area: Which design decision is better: setting the same W and L for the mirror and cascode devices? Or using larger W and L for the current mirror devices? Why?

The best decision is to use large area (large W and L) for the **current mirror** devices to reduce the mismatch error, **also** we can optimize the area by **reducing the area** of the **cascode** device since they have very small percent change in the output current.

Monto carlo simulation



Simple CM Histogram for 200 MC runs



Wide swing CM Histogram for 200 MC runs

	A	B	C	D	E	F	G
1	@m.xm1.m0[id]	=	2.01E-05		average	sigma	%error
2	@m.xm1.m0[id]	=	2.00E-05		2.01E-05	3.87175E-07	1.93E+00
3	@m.xm1.m0[id]	=	1.99E-05				
4	@m.xm1.m0[id]	=	1.93E-05				
5	@m.xm1.m0[id]	=	1.97E-05				

Simple CM results

	A	B	C	D	E	F	G
1	@m.xm3.m0[id]	=	1.96E-05		avg	sigma	%error
2	@m.xm3.m0[id]	=	1.99E-05		2.01E-05	3.7575E-07	1.87E+00
3	@m.xm3.m0[id]	=	1.96E-05				
4	@m.xm3.m0[id]	=	1.96E-05				
5	@m.xm3.m0[id]	=	2.02E-05				

Wide swing CM results

parameter	Simulation	Analytical	Error
Wide swing CM %error	1.87%	2%	0.13
Simple CM %error	1.93%	2%	0.07